

USER ACCEPTANCE TESTING

RDHIS2 Mobile Tracker

UAT Test Checklist — Version 1.0.0

Prepared for the Ministry of Health / HISP Ethiopia UAT session · 29 September 2026

TESTER NAME	TEST DATE
DEVICE / OS	BUILD / APK VERSION
DHIS2 SERVER URL	TEST ACCOUNT / ORG UNIT
NETWORK CONDITIONS TESTED	TOTAL TEST CASES (155)

Purpose. This checklist walks a tester through every user-facing feature of the RDHIS2 Mobile Tracker. Each row is a single test case with the steps to perform and the expected result. Mark **Pass** or **Fail**, and record the actual behaviour, screenshots or defect IDs in the Notes column.

How to use it.

- Work top to bottom within each section; sections are independent, so you may split them across testers.
- A case passes only when the observed result matches the expected result in full.
- For any fail, capture a screenshot, note the exact on-screen text and log a defect ID.
- Re-test offline-first behaviour explicitly (airplane mode) — it is central to this app.
- Use a **staging/test** DHIS2 server and non-production accounts. Do not enter real patient data.

Legend:

☐ Pass

☐ Fail

N/A — note why in Notes

Contents	
A	Onboarding & First-Run Experience 9
B	Authentication & Session 13
C	Home Shell & Navigation 12
D	Capture Workflow (Org Unit -> Dataset -> Section -> Period) 13
E	Reports Overview & To-do Band 10
F	Data Entry Form 21
G	Data Quality & Validation 14
H	Offline-First & Synchronization 17
I	Visualization & Dashboards 15
J	Export (PDF & Excel) 5
K	Notifications & Deadline Reminders 6
L	Settings 4
M	Edge Cases & Error Handling 9
N	Non-Functional Checks 7

A Onboarding & First-Run Experience				
Fresh-install behaviour, the welcome carousel, spotlight app tours and the one-time battery-optimisation prompt.				
ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
ONB-01	First-launch carousel Fresh install. Open the app for the very first time.	Four slides appear in order: 'Welcome to RDHIS2 Mobile App', 'Capture data anywhere', 'Syncs when you're connected', 'Build and view charts'. Each has an illustration, title and body text.	☐ Pass ☐ Fail	
ONB-02	Carousel navigation Tap 'Next' through the slides; on the last slide read the button.	Button reads 'Next' on slides 1-3 and changes to 'Get Started' on the last. 'Skip' is visible on slides 1-3 and hidden on the last.	☐ Pass ☐ Fail	
ONB-03	Skip onboarding On slide 1 tap 'Skip'.	Carousel closes and the Login screen is shown. Relaunching the app does not show the carousel again.	☐ Pass ☐ Fail	
ONB-04	Complete onboarding On the last slide tap 'Get Started'.	Carousel closes and the Login screen is shown.	☐ Pass ☐ Fail	
ONB-05	Home spotlight tour Log in and land on Home for the first time.	A spotlight tour highlights the menu, search, sync button and the Visualization/Capture toggle ('Switch modes'). It can be stepped through.	☐ Pass ☐ Fail	
ONB-06	Per-screen tours Visit a routine form, a disease-registration form and the Visualization tab.	Each screen shows its own one-time spotlight tour (routine entry, disease entry, dashboards/builder). They do not repeat on later visits.	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
ONB-07	Replay the app tour Open the drawer and tap 'App Tour'; return to Home, then revisit the other screens.	Home's tour replays immediately; the other screens replay on their next visit.	☐ Pass ☐ Fail	
ONB-08	Battery-optimisation prompt Log in on an Android device the first time.	Dialog 'Keep offline sync reliable' appears with 'Not now' / 'Allow'. Choosing either dismisses it and it is never shown again.	☐ Pass ☐ Fail	
ONB-09	Prompt fail-open Dismiss the battery prompt and continue working offline.	No repeated nagging; the app continues to function if the prompt errors.	☐ Pass ☐ Fail	

B Authentication & Session

Online and offline login, credential validation, server switching, session expiry and logout.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
AUTH-01	Online login with valid credentials On Wi-Fi/data, enter a valid username and password, tap 'Log in'.	Login succeeds and Home opens. On the very first login a 'Downloading data' overlay appears while metadata is pulled for offline use.	☐ Pass ☐ Fail	
AUTH-02	Invalid credentials Enter a wrong username or password while online and submit.	Login is refused with a clear message such as 'Invalid server URL or credentials.' The user stays on the login screen.	☐ Pass ☐ Fail	
AUTH-03	Empty-field validation Leave username and/or password blank and submit.	'Please enter your username' and/or 'Please enter your password' is shown.	☐ Pass ☐ Fail	
AUTH-04	Short-password validation Enter a password shorter than the minimum and submit.	'Password is too short' is shown; no login attempt is made.	☐ Pass ☐ Fail	
AUTH-05	Offline login (previously used account) Log in online once, log out, enable airplane mode, log in with the same credentials.	Login succeeds offline using the locally stored verifier. No network error is shown.	☐ Pass ☐ Fail	
AUTH-06	Offline login (never-used account) On a fresh device in airplane mode, try an account never logged in before.	Login fails with a clear message; the app does not hang.	☐ Pass ☐ Fail	
AUTH-07	Change DHIS2 server URL Tap the server icon on the login screen; open 'DHIS2 Server'.	Dialog shows the hint 'https://server/api'. An invalid address shows 'Enter a valid server address'. A valid address save shows 'Server set to <url>'.	☐ Pass ☐ Fail	
AUTH-08	Session expiry / 401 Invalidate the session server-side (or wait for expiry), then use the app.	The user is returned to login with 'Your session has ended. Please log in again.'	☐ Pass ☐ Fail	
AUTH-09	Logout from drawer Drawer -> 'Log Out'.	App returns to the Login screen.	☐ Pass ☐ Fail	
AUTH-10	Logout from Settings (confirmation) Settings -> 'Log out'.	Confirm dialog 'Log out?' with text 'You will need to log in again. Data not yet synced stays on this device.', plus 'Cancel' / 'Log out'.	☐ Pass ☐ Fail	
AUTH-11	Re-login preserves local data With draft/unsynced values on the device, log out and log back in as the same user.	The drafts and unsynced values are still present; the same local database is reused.	☐ Pass ☐ Fail	
AUTH-12	Login pushes queued work Create offline pending data, then log out and log in while online.	Queued values are pushed automatically as part of login.	☐ Pass ☐ Fail	
AUTH-13	Per-user data isolation Log in as user A, enter data, log out, log in as user B.	User B cannot see user A's drafts or data; each user has a separate database.	☐ Pass ☐ Fail	

C Home Shell & Navigation

The mode toggle, mode-aware search, app-bar actions, filters, drawer and connectivity indicator.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
HOME-01	Default mode Open Home after login.	Visualization mode is selected by default.	☐ Pass ☐ Fail	
HOME-02	Mode toggle Tap between 'Visualization' and 'Capture'.	The content area swaps between dashboards and the organisation-unit tree.	☐ Pass ☐ Fail	
HOME-03	Mode-aware search hint Open search in each mode.	Capture mode hint: 'Search reports...'. Visualization mode hint: 'Search dashboards or charts...'.	☐ Pass ☐ Fail	
HOME-04	Drawer items Open the drawer.	Items present: Home, Setting, App Tour, Log Out, About. Each opens the expected destination.	☐ Pass ☐ Fail	
HOME-05	About dialog Drawer -> 'About'.	Shows app name and 'Version 1.0.0' plus description; 'Close' dismisses it.	☐ Pass ☐ Fail	
HOME-06	Connectivity indicator Toggle airplane mode on/off while on Home.	Indicator shows 'Online' / 'Offline' with tooltips 'Connected to the server' / 'No server connection — changes are saved on this device'.	☐ Pass ☐ Fail	
HOME-07	Sync button states Observe the sync button before/during/after a sync.	Tooltips read 'Sync' / 'Syncing' / 'Sync all'; a result snackbar is shown when done.	☐ Pass ☐ Fail	
HOME-08	Filter panel rows Switch to Capture and open filters.	Rows labelled DATE, ORG. UNIT, SYNC. Idle summary reads 'No filters applied'.	☐ Pass ☐ Fail	
HOME-09	Date filter options Open the DATE filter.	Options include Today, Yesterday, Tomorrow, This/Last/Next Week, This/Last/Next month, From -To, Other, Any time. A custom range shows as dd/MM/yyyy.	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
HOME-10	Sync-state filter options Open the SYNC filter.	Options are 'Synced', 'UnSynced', 'Sync Error'. Selecting one filters the report list.	☐ Pass ☐ Fail	
HOME-11	Organisation-unit filter Open ORG. UNIT; search and use 'Choose from organisation unit tree'.	Search filters the list; the full-page tree allows selection; 'Apply Filter' applies and a clear control removes it.	☐ Pass ☐ Fail	
HOME-12	Sync refreshes capture tree In Capture with unsynced chips shown, tap Sync.	On completion the tree/chips refresh to reflect the just-completed push.	☐ Pass ☐ Fail	

D Capture Workflow (Org Unit -> Dataset -> Section -> Period)

The offline capture navigation: lazily loaded org-unit tree, dataset selection, sections, the Ethiopian-calendar period picker and disease-registration combos.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
CAP-01	Org-unit tree lazy loading Open Capture; expand nodes one level at a time.	Only one level loads per expand ('Loading organisation units...' while fetching). The full ~38,000-unit tree is never loaded at once and the UI stays responsive.	☐ Pass ☐ Fail	
CAP-02	Org-unit search & empty states Search for an org unit; also test a query with no matches.	Matching units are listed; a no-match query shows 'No results for "<query>". Header reads 'Select an organisation unit to capture data for'.	☐ Pass ☐ Fail	
CAP-03	Select an organisation unit Tap an org unit leaf.	Header shows 'Selected: <name>' and 'Continue' is enabled to proceed.	☐ Pass ☐ Fail	
CAP-04	Dataset list Open 'Select Dataset' for an org unit.	Datasets assigned to the org unit are listed, each searchable, with a Synced/Unsynced chip. Empty state: 'No datasets available' or the 'No datasets are assigned to <orgUnit>' message.	☐ Pass ☐ Fail	
CAP-05	Load-failure state Open the dataset list with the local DB empty / metadata missing.	A 'Could not load datasets' error with 'Try Again' (or a prompt to log in online) is shown; retry recovers.	☐ Pass ☐ Fail	
CAP-06	Dataset 'Sync all' From the dataset page tap 'Sync all' while online.	Button changes to 'Syncing' and the list refreshes after the push.	☐ Pass ☐ Fail	
CAP-07	Section selection Open a dataset that has sections.	Page 'Select Section' lists numbered sections; subtitle shows '<dataset> · <period>'. Datasets without sections skip straight to the period.	☐ Pass ☐ Fail	
CAP-08	Period picker Reach 'Report Period'.	Field labelled 'Report Period', defaults to the current period for the dataset's period type (default 'Monthly'). Ethiopian month names are displayed.	☐ Pass ☐ Fail	
CAP-09	Period gating Try to select an expired or not-yet-open period.	Expired periods are not selectable; future ones may show 'not open yet'. Period open/expired honours expiryDays and openFuturePeriods.	☐ Pass ☐ Fail	
CAP-10	Clock-tamper warning With the tamper flag set, open the period picker.	Warning shown: "This device's clock was set backwards. Only the current period is open until the app reconnects to the server."	☐ Pass ☐ Fail	
CAP-11	Disease registration flow Open a Disease Registration dataset.	A category-combo picker appears before the form; the app bar/section show 'Disease Registration' and the form is themed accordingly.	☐ Pass ☐ Fail	
CAP-12	New-report entry point Tap the '+' / 'New report' button.	The capture flow starts (org unit -> dataset -> ...) as expected.	☐ Pass ☐ Fail	
CAP-13	Category-combo resolution error Select combos when metadata is incomplete.	Banner: 'Could not resolve the selected combination — try again once metadata has synced.' The user can recover.	☐ Pass ☐ Fail	

E Reports Overview & To-do Band

The Report Period list (all reports across all org units) and the 'Reports to fill' / To-do band with urgency.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
RPT-01	Empty report list Open Report Period with no reports touched.	'No reports yet' plus guidance to tap + to start one.	☐ Pass ☐ Fail	
RPT-02	Reports across all org units Complete/draft reports at more than one facility.	All reports appear in one list, routine and disease registration merged.	☐ Pass ☐ Fail	
RPT-03	Report status chips Inspect report cards.	Cards show 'Completed'/'Incomplete' and 'Synced'/'Unsynced' as applicable.	☐ Pass ☐ Fail	
RPT-04	Report card identity Inspect a disease-registration report card.	Card reads '<period> · <org unit>' and, for disease registration, also '<combo>' so reports sharing dataset/period/org unit are not conflated.	☐ Pass ☐ Fail	
RPT-05	'Reports to fill' band Open the To-do band online and offline.	Lists reports still owed. Empty state: 'Nothing to fill' + 'Every report assigned to your organisation units is up to date.'	☐ Pass ☐ Fail	
RPT-06	Urgency labels Inspect items due at different times.	Labels show Overdue / Due soon / Open with the relevant lock date (e.g. 'Overdue — locked 5 Jan', 'Locks 5 Jan').	☐ Pass ☐ Fail	
RPT-07	Started / Not started Inspect an item that has local work vs one with none.	'Started' is shown when something is already queued; otherwise 'Not started'.	☐ Pass ☐ Fail	
RPT-08	Report search & filters Search and filter the report list with no matches.	Matching reports remain; a no-match state shows 'No results' + 'No reports match the current search or filters.'	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
RPT-09	Reopening un-finishes a report Open a completed report and save a draft instead of completing.	The report shows as 'Incomplete' in the list (intended behaviour).	☐ Pass ☐ Fail	
RPT-10	Server completion pull Complete a report for the same dataset/period/org unit on the web, then refresh in-app.	The item disappears from 'Reports to fill' after the best-effort server pull.	☐ Pass ☐ Fail	

F Data Entry Form

Opening, editing, saving, completing and reopening forms, plus all the entry controls (options, booleans, controller elements, disease list, totals).

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
DE-01	Open a form online Open a routine form while online.	Server values for that form are pulled, conflicts resolved, and the form renders. A pull failure does not block the form.	☐ Pass ☐ Fail	
DE-02	Open a form offline Open the same form in airplane mode after a prior sync.	The form opens from local data with cached values; no blocking network error.	☐ Pass ☐ Fail	
DE-03	Edit a numeric cell Type a number into a numeric/ integer/ positive-integer cell.	The value is accepted and shown; the cell is marked modified.	☐ Pass ☐ Fail	
DE-04	Edit other value types Enter values into percentage, unit-interval, date, phone and e-mail cells.	Each type accepts only valid input; invalid input is flagged inline.	☐ Pass ☐ Fail	
DE-05	Option-set picker Tap a cell that uses an option set.	An option grid opens; choosing an option sets the value. 'Clear value' removes it.	☐ Pass ☐ Fail	
DE-06	Boolean / true-only cells Tap boolean and true-only cells repeatedly.	Values cycle 'Yes'/'No'/'-' (true-only toggles Yes/none).	☐ Pass ☐ Fail	
DE-07	Greyed-out field Hover/tap a greyed field.	Not editable; tooltip "This combination isn't applicable and can't be entered."	☐ Pass ☐ Fail	
DE-08	Controller element gating Change a controller element that others depend on from Yes to No.	Confirm dialog 'Clear entered data?' explains N values will be erased; 'Clear and close' erases and hides them, 'Cancel' reverts.	☐ Pass ☐ Fail	
DE-09	Save as draft Edit values and tap the 'Save' FAB.	Values save locally as a draft; snackbar confirms saving as a draft on this device.	☐ Pass ☐ Fail	
DE-10	Invalid value blocks save Enter an invalid value (e.g. letters in a number cell) and tap Save.	Save is blocked; message names the first problem and asks to fix the red cell(s). Nothing invalid reaches the database.	☐ Pass ☐ Fail	
DE-11	Complete - clean form With all mandatory fields filled and no issues, tap Save then 'Complete'.	Dialog 'Everything looks good'; completing shows 'Data set completed successfully!' and values are queued/pushed.	☐ Pass ☐ Fail	
DE-12	Complete - missing mandatory fields Leave a compulsory field blank and try to complete.	Dialog 'N required field(s) missing' blocks completion; the required cell is indicated.	☐ Pass ☐ Fail	
DE-13	Complete - validation issues Create a validation-rule violation and try to complete.	Dialog 'N validation issue(s) found' warns; the user may 'Review data' or 'Complete anyway'.	☐ Pass ☐ Fail	
DE-14	Reopen a completed form Open a completed report, change a value and tap Save.	Dialog 'This data set is completed' offers 'Incomplete' (reopen, edits stay drafts) or 'Keep completed' (sends saved changes and stays completed).	☐ Pass ☐ Fail	
DE-15	Closed period is view-only Open a form whose period is closed.	Header chip 'Closed · View only'; the Save FAB is hidden and cells are read-only.	☐ Pass ☐ Fail	
DE-16	Search within the form Use the form search field.	Data elements filter as typed; 'No results for "<query>"' when empty.	☐ Pass ☐ Fail	
DE-17	Reload values Make an unsaved edit, then tap 'Reload values'.	If unsaved changes exist, 'Save your changes before reloading.'; otherwise server/local values reload.	☐ Pass ☐ Fail	
DE-18	Row totals & calculated fields Inspect summation rows and auto-calculated cells.	Totals sum the row; calculated cells show 'Automatically calculated — not entered directly'.	☐ Pass ☐ Fail	
DE-19	Previous-entries hint Enter a numeric value in a cell with history.	Hint shows 'Higher/Lower/Within last N' with the previous values listed.	☐ Pass ☐ Fail	
DE-20	Disease entry list On a disease form, use 'Select for new disease'.	A new disease form is added; existing diseases can be 'Collapse'/'Expand'; when all are added, 'Every disease has already been added'.	☐ Pass ☐ Fail	
DE-21	Section navigation Move between collapsible sections in a large form.	Sections expand/collapse smoothly and the correct fields are shown.	☐ Pass ☐ Fail	

G Data Quality & Validation

The offline data-quality suite: value-type checks, validation rules, mandatory fields, outliers and the per-cell audit trail.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
DQ-01	Value-type validation Type an out-of-range or wrong-format value into a typed cell.	The cell is flagged and the value cannot be saved with the form.	☐ Pass ☐ Fail	
DQ-02	Validation rules warn (do not block) Enter values that violate a synced DHIS2 validation rule.	A warning is shown at completion and in the live banner; completing anyway is still possible (same contract as the DHIS2 web app).	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
DQ-03	Live validation banner Enter values that trigger a rule while typing.	Banner 'N validation issue(s) — tap to review' updates live; tapping shows details with rule name and comparison.	☐ Pass ☐ Fail	
DQ-04	Mandatory fields block completion Attempt to complete with compulsory operands blank.	Completion is blocked until the mandatory field(s) are filled.	☐ Pass ☐ Fail	
DQ-05	Grey fields cannot be entered Try to edit a server-configured grey field.	Field is non-editable.	☐ Pass ☐ Fail	
DQ-06	Controller element hides group Set a controller element to No.	Dependent elements are hidden and their existing values cleared after confirmation.	☐ Pass ☐ Fail	
DQ-07	Outlier warning (single value) Type a value far outside the cell's recent history.	After ~600 ms a dialog 'Are you sure?' shows the typical value, historical range and bounds, with 'Go back and correct' / 'Save value'.	☐ Pass ☐ Fail	
DQ-08	Outlier acknowledgement Choose 'Save value' on an outlier, then edit the cell again.	The value is kept and the warning is not repeated for that cell until settings change.	☐ Pass ☐ Fail	
DQ-09	Outlier warning (multiple) Produce several outliers at once.	Dialog lists the values with 'N values look very different...' and offers 'Go back' / 'Save anyway'.	☐ Pass ☐ Fail	
DQ-10	Outlier settings Open 'Outlier check' from the form menu.	Algorithms 'Z-score', 'Modified Z-score', 'Min-Max' and a threshold slider ('more sensitive' ... 'less sensitive') are available; 'Save' applies and re-runs the check.	☐ Pass ☐ Fail	
DQ-11	Audit trail - local edits Long-press a cell with local edits and open its history.	History sheet shows 'On this device' entries with old/new value, actor and time (statuses include 'Saved as a draft', 'Waiting to sync', 'Synced', 'Rejected').	☐ Pass ☐ Fail	
DQ-12	Audit trail - server history Open history online for a cell changed on another client.	Server entries appear (CREATE/UPDATE/DELETE) under the server banner 'Change history below is from the DHIS2 server'.	☐ Pass ☐ Fail	
DQ-13	Audit trail - offline Open history in airplane mode.	Banner 'Offline — showing this device's own edit log below'; local entries still show.	☐ Pass ☐ Fail	
DQ-14	Audit trail - empty Open history for a never-edited cell.	'No changes recorded for this cell yet.'	☐ Pass ☐ Fail	

H Offline-First & Synchronization

Draft -> pending -> synced/error state machine, automatic and manual sync, conflict resolution, server rejections and metadata sync.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
SYNC-01	Full offline data entry In airplane mode: browse, open a form, enter values, save and complete.	Everything works with no network. Nothing is lost; values are queued locally.	☐ Pass ☐ Fail	
SYNC-02	State transitions Observe a value's state after complete and after a successful push.	Value moves draft -> pending on Complete -> synced after the server accepts it.	☐ Pass ☐ Fail	
SYNC-03	Auto-sync on reconnect Create offline pending data, then restore connectivity.	Queued values upload automatically on the offline->online transition.	☐ Pass ☐ Fail	
SYNC-04	Auto-sync on login Log in while online with pending data already on the device.	Queued values upload as part of login.	☐ Pass ☐ Fail	
SYNC-05	Heartbeat sync Leave pending data and stay online without any event.	Within the 5-minute heartbeat the pending values are eventually pushed.	☐ Pass ☐ Fail	
SYNC-06	Manual sync outcomes Tap the sync button in each situation: offline with pending, offline with none, all synced, uploaded, partial, failed.	Each shows its specific message, e.g. 'Sync complete — N entry/entries uploaded.', 'Everything is already synced.', 'Sync failed — your entries are safe on this device...'	☐ Pass ☐ Fail	
SYNC-07	Conflict - draft always wins Create a draft, change the same cell on the web, reopen the form online.	The local draft is kept and not overwritten by the server value.	☐ Pass ☐ Fail	
SYNC-08	Conflict - newest wins With a pending value differing from the server, reopen the form online.	The newer of the two wins; a genuinely newer local edit stays pending and is pushed.	☐ Pass ☐ Fail	
SYNC-09	Conflict - tampered clock Set the device clock back, create a conflicting pending value, reopen online.	The server value wins while the clock is suspect.	☐ Pass ☐ Fail	
SYNC-10	Server rejection handling Push a value the server rejects (wrong type / not in option set).	The cell turns red with the server's reason; editing it clears the error and re-queues it as pending.	☐ Pass ☐ Fail	
SYNC-11	Completion registration push Complete a dataset online, then mark it incomplete.	Completion is POSTed (and DELETE) to the server; reopen succeeds.	☐ Pass ☐ Fail	
SYNC-12	Metadata full sync Log in for the first time on a fresh device.	A full metadata download scoped to the user's own org-unit subtree completes; only then is the device ready for offline capture.	☐ Pass ☐ Fail	
SYNC-13	Metadata delta sync Log in again online later.	Only changed/new metadata is fetched; deletions on the server are removed locally.	☐ Pass ☐ Fail	
SYNC-14	Offline org-unit depth bound Inspect the offline org-unit tree for a woreda-assigned user.	Capture roots plus their direct children are stored; deeper levels remain reachable online via the live fallback.	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
SYNC-15	Tamper-resistant clock Set the device clock backwards, then reopen the app.	Past periods are refused/warned; the current period still works; reconnecting anchors the clock to the server and clears the flag.	☐ Pass ☐ Fail	
SYNC-16	Background kill protection (Android) Start a sync, background the app and check notifications.	A foreground 'dataSync' notification appears only while work is in flight; if the process is killed, pending values remain queued and retry later.	☐ Pass ☐ Fail	
SYNC-17	Transport failure safety Cause a sync to fail mid-flight (drop network during push).	Values remain pending (never marked synced); a later sync retries them.	☐ Pass ☐ Fail	

I Visualization & Dashboards

Server dashboards, on-device local dashboards, the chart builder and offline caching.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
VIS-01	Three visualization tabs Open the Visualization tab.	Flat tabs 'Server Dashboard', 'Local Dashboard', 'Create New' are present.	☐ Pass ☐ Fail	
VIS-02	Server dashboards load Open 'Server Dashboard' online.	DHIS2 server dashboards are listed; opening one renders its charts natively. Each chart loads independently, so one failure does not block the rest.	☐ Pass ☐ Fail	
VIS-03	Per-chart retry With a chart that failed to load, tap its 'Retry'.	Only that chart re-fetches; the rest of the dashboard is unaffected.	☐ Pass ☐ Fail	
VIS-04	Unsupported items notice Open a dashboard containing unsupported item types (e.g. maps).	A notice such as 'N items not supported' is shown; the rest renders.	☐ Pass ☐ Fail	
VIS-05	Offline cache banner Load a dashboard, then switch to airplane mode and reopen it.	Charts render from cache with 'Offline — showing cached data' (and a timestamp).	☐ Pass ☐ Fail	
VIS-06	Local dashboard empty state Open 'Local Dashboard' with no saved charts.	'No local charts yet' + 'Build your first chart...'	☐ Pass ☐ Fail	
VIS-07	Create a chart In 'Create New', choose chart type, data type, group/items, org unit and period, then save.	'Chart saved.'; the chart appears under Local Dashboard and is never pushed to the server.	☐ Pass ☐ Fail	
VIS-08	Chart types render Create/view Column, Bar, Line, Pie, Single Value and Gauge charts.	Each renders correctly; an empty result shows 'No data'.	☐ Pass ☐ Fail	
VIS-09	Data types Build charts from Indicator, Data Element and Dataset sources.	Each source offers the right group/item picker and metric options (Reporting Rate, Actual Reports, etc.).	☐ Pass ☐ Fail	
VIS-10	Relative vs fixed period Toggle 'Relative Period' / 'Fixed Period' in the builder.	Relative flags (This Month, Last 3 Months, This Financial Year, ...) and fixed periods both produce a query.	☐ Pass ☐ Fail	
VIS-11	Edit a local chart Open a saved chart and tap 'Edit chart'.	'Edit Chart' page loads; 'Save Changes' updates it and shows 'Chart updated.'	☐ Pass ☐ Fail	
VIS-12	Delete a local chart Tap the delete action on a local chart.	Dialog 'Delete chart?' with '<name>' will be removed from your charts.'; 'Delete' removes it.	☐ Pass ☐ Fail	
VIS-13	Offline chart-builder draft Build a chart while offline and tap 'Save as Draft'.	A 'Saved offline — will finish building once back online' chip appears; on regaining connectivity the draft is finished automatically.	☐ Pass ☐ Fail	
VIS-14	Misconfigured visualization Attempt to render a visualization with no period dimension.	A clear, displayable configuration message is shown instead of a generic network error.	☐ Pass ☐ Fail	
VIS-15	Search dashboards/charts Search in Visualization mode.	Dashboards are searched in Server mode and local charts in Local mode; no-match states are shown.	☐ Pass ☐ Fail	

J Export (PDF & Excel)

Exporting an open form for paper backup or off-device review.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
EXP-01	Export as PDF Open a form, tap 'Download' -> 'Download as PDF'.	Prompt 'What should the PDF include?' with 'Only recorded' / 'All elements'. A saved/shared PDF is produced.	☐ Pass ☐ Fail	
EXP-02	PDF content Open the exported PDF.	Header '<org unit> · <period>', 'Generated <date>', a value table, 'Page N / M' footers, and 'Nothing to show.' when empty.	☐ Pass ☐ Fail	
EXP-03	Export as Excel Open a form, tap 'Download' -> 'Download as Excel'.	An .xlsx is shared with a 'Data' sheet, columns 'Data Element' / 'Category' / 'Value', and a 'Generated by <user>' line.	☐ Pass ☐ Fail	
EXP-04	Export offline Export a form in airplane mode.	Export succeeds from local data.	☐ Pass ☐ Fail	
EXP-05	Export filenames Check the suggested filename for each export.	Format '<title> - <period>.pdf' / '.xlsx'.	☐ Pass ☐ Fail	

K Notifications & Deadline Reminders

On-device local reminders for reports due before their lock date.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
NOT-01	Reminder permission First time reminders are needed, grant/deny notification permission.	Permission is requested; if denied the app disables reminders silently and keeps working.	☐ Pass ☐ Fail	
NOT-02	3-day heads-up Have a report locking in 3 days and wait/trigger the schedule.	At 08:00 (Africa/Addis_Ababa) a 'Reports lock soon' notification lists the report(s).	☐ Pass ☐ Fail	
NOT-03	Lock-day reminder Have a report locking today.	A 'Reports lock today' notification is delivered at 08:00.	☐ Pass ☐ Fail	
NOT-04	Grouping by facility/date Have multiple reports locking on the same day at one facility.	A single grouped notification is delivered, not one per report.	☐ Pass ☐ Fail	
NOT-05	Reschedule on change Complete one of the pending reports and refresh the To-do list.	The stale reminder is cancelled and the schedule reflects the remaining reports.	☐ Pass ☐ Fail	
NOT-06	Reboot behaviour (Android) Reboot the device.	Reminders are re-armed (or rescheduled on next launch) without duplication.	☐ Pass ☐ Fail	

L Settings

Account, server and app information plus logout.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
SET-01	Profile information Open Settings.	Shows the logged-in username and primary organisation unit (fallbacks 'Not logged in' / 'No organisation unit').	☐ Pass ☐ Fail	
SET-02	Server URL display/edit Inspect and change the server URL from Settings.	Current URL is shown; 'DHIS2 Server' dialog validates and saves it.	☐ Pass ☐ Fail	
SET-03	App version Inspect the About section.	Shows 'App version' with 'RDHIS2 Mobile App 1.0.0'.	☐ Pass ☐ Fail	
SET-04	Logout confirmation Tap 'Log out'.	Dialog 'Log out?' with the 'Data not yet synced stays on this device.' notice; confirming returns to login.	☐ Pass ☐ Fail	

M Edge Cases & Error Handling

Recovery paths and boundary conditions testers should actively try to break.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
EDGE-01	No connectivity at first use Fresh install, no network, attempt login.	A clear 'No internet connection.'-style message is shown; the app does not crash.	☐ Pass ☐ Fail	
EDGE-02	Server error / timeout Point at an unreachable or erroring server and act.	'Server error. Please try again.' / 'Connection timed out. Please try again.' messages are shown; local data is unaffected.	☐ Pass ☐ Fail	
EDGE-03	No datasets assigned Open an org unit with no assigned datasets.	Guidance is shown: 'No datasets are assigned to <orgUnit>...' to pick another unit or contact an administrator.	☐ Pass ☐ Fail	
EDGE-04	Same report, different combos Create two disease reports with the same dataset/period/org unit but different combos.	They remain separate entries and are not conflated.	☐ Pass ☐ Fail	
EDGE-05	Large org-unit hierarchy performance Expand deep branches of the national org-unit tree.	The tree stays responsive (lazy per-level loading); no freeze or memory spike.	☐ Pass ☐ Fail	
EDGE-06	App backgrounded during entry Enter values, background the app, return.	Entered values are preserved and the form/state is intact.	☐ Pass ☐ Fail	
EDGE-07	Rotation / small screen Rotate the device and test on a small screen.	Layouts reflow without clipping key controls.	☐ Pass ☐ Fail	
EDGE-08	Duplicate / rapid taps Rapidly tap Save and Complete.	No duplicate submissions or crashes; state remains consistent.	☐ Pass ☐ Fail	
EDGE-09	Storage pressure Fill the local database with many values and sync.	Batches are chunked (500 values/request) and no data is lost on a mid-transfer drop.	☐ Pass ☐ Fail	

N Non-Functional Checks

Performance, reliability, security and usability observations.

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
NFR-01	Performance Time form open, org-unit expansion, saving and syncing.	All interactions complete in an acceptable time with no visible jank.	☐ Pass ☐ Fail	
NFR-02	Battery / background behaviour Leave the app installed and connected over a period.	Heartbeat and reminders do not noticeably drain the battery.	☐ Pass ☐ Fail	
NFR-03	Credential security Inspect stored credentials.	Passwords are not stored in plain text (a salted verifier is used); session data is in secure storage.	☐ Pass ☐ Fail	
NFR-04	Data durability Force-kill the app mid-sync.	No committed data is lost; pending items retry on next launch.	☐ Pass ☐ Fail	
NFR-05	Text scaling / accessibility Increase the system font size.	Labels remain readable and controls usable.	☐ Pass ☐ Fail	
NFR-06	Localization Inspect the UI language.	UI is English with the Amharic Ministry heading on login; note any untranslated text.	☐ Pass ☐ Fail	

ID	Test Case & Steps	Expected Result	Result	Notes / Evidence
NFR-07	Release HTTPS-only In a release build, configure an http:// server.	The insecure server is refused (http works only in debug builds).	☐ Pass ☐ Fail	

Defect Log

Defect ID	Test Case ID	Description / Steps to reproduce	Severity	Screenshot / Attachment

Overall Result

☐ Accepted — ready to release ☐ Accepted with minor issues ☐ Rejected — must re-test

Tester signature / name

Date

Development / QA representative

Date